Question ID: e4b4e9ea

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The length of the edge of the base of a right square prism is 6 units. The volume of the prism is $2{,}880$ cubic units. What is the height, in units, of the prism?

Answer

A. $4\sqrt{30}$

B. 36

C. $24\sqrt{5}$

D. 80

Correct Answer: D

Rationale

Choice D is correct. The volume, V, of a right square prism is given by the formula $V = s^2h$, where s represents the length of the edge of the base and h represents the height of the prism. It's given that the volume of a right square prism is $2{,}880$ cubic units and the length of the edge of the base is 6 units. Substituting $2{,}880$ for V and 6 for s in the formula $V = s^2h$ yields $2{,}880 = (6^2)h$, or $2{,}880 = 36h$. Dividing both sides of this equation by 36 yields $80 = h$. Therefore, the height, in units, of the prism is 80.

Choice A is incorrect. This is the height, in units, of a right square prism where the length of the edge of the base is 6 units and the volume of the prism is $144\sqrt{30}$, not $2{,}880$, units.

Choice B is incorrect. This is the area, in square units, of the base, not the height, in units, of the prism.

Choice C is incorrect. This is the height, in units, of a right square prism where the length of the edge of the base is 6 units and the volume of the prism is $864\sqrt{5}$, not $2{,}880$, units.